# AI and Data Scientist

## Mathematics

- [ ] Linear Algebra, Calculus, Mathematical Analysis
  - Mathematics for Machine Learning
- [ ] Dilerential Calculus
  - Coursera: Algebra and Dilerential Calculus

## Statistics

- [ ] Statistics, CLT
  - Coursera: Introduction to Statistics
- [ ] Hypothesis Testing
  - Coursera: Hypothesis Testing
- [ ] Probability and Sampling
  - Coursera: Probability and Statistics
- [ ] AB Testing
  - Practitioner's Guide to Statistical Tests
  - Experiment Design Article
- [ ] Increasing Test Sensitivity
  - Minimum Detectable Elect
  - Paper: Improving Test Sensitivity
  - Paper: Improving Sensitivity (CUPED)
  - CUPED at Booking.com
  - Doordash: CUPAC
  - Netflix: Stratification
- [ ] Ratio Metrics
  - Microsoft: Delta Method in Metric Analytics
  - Paper: Ratio Metrics

## Econometrics

- [ ] Pre-requisites of Econometrics
  - Fundamentals of econometrics
- [ ] Regressions, time series, fitting distributions
  - Intro to Econometrics
  - Coursera: Econometrics
  - Kaggle — Learn Time Series
  - Kaggle — Time Series Basics
  - ARIMA Model for Time Series
  - Time Series Models
  - Forecasting Task with Solution
  - Coursera: Linear Regression

## Coding

- [ ] Learn Python Programming Language
  - Learn Python - Kaggle
  - Google's Python Class
- [ ] Data Structures and Algorithms (Python)
  - Algorithmic Exercises
  - Study Plans - Leetcode
  - Algorithms Specialization
- [ ] Learn SQL
  - SQL Tutorial

## Exploratory Data Analysis (EDA)

- [ ] Data understanding, Data Analysis, Vizualisation
  - Exploratory Data Analysis With Python and Pandas
  - Exploratory Data Analysis for Machine Learning
  - Exploratory Data Analysis with Seaborn

## Machine Learning

- [ ] Classic ML (Sup. and Unsup.) Advanced ML (Ensembles, NNs)
  - Open Machine Learning Course - OpenDataScience
  - Machine Learning Specialization
  - Pattern Recognition & ML by Christopher m. bishop
  - GitHub Repository with Notes and Code from the eBook Above

## Deep Learning

- [ ] Fully connected NN, CNN, RNN, LSTM, Transformers, Transfer Learning
  - Deep Learning Specialization
  - Deep Learning Book
  - Attention is all you need
  - The Illustrated Transformer

## MLOps

- [ ] Deployment models, CI/CD
  - MLOps Specialization

Akash • Python & Tech Enthusiast 🚀
@pycode.hubb